NextUp Enterprise Platform Architecture Guide

Mobile Application vs Web Portal Comparison & General Manager Executive Authorities

System Version: v2.3.0 (Build 15) â¢ Prepared for Rathinam Group of Institutions â¢ November 2026

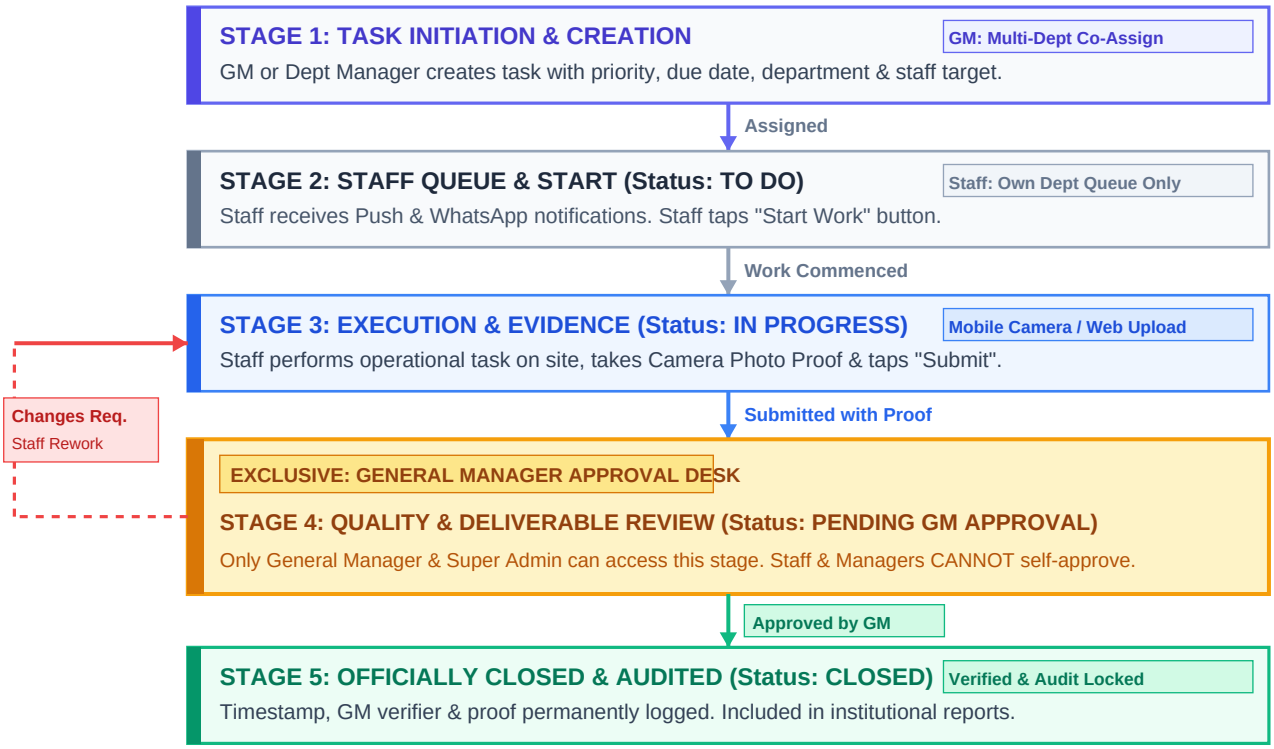
EXECUTIVE PLATFORM SUMMARY

NextUp is engineered as a dual-surface operational platform. The Web Portal provides comprehensive institutional governance, multi-view workspaces, and staff management. The Mobile Application provides high-speed field operations, shift checklists, live photo deliverables, and instant WhatsApp dispatch. General Managers hold exclusive cross-departmental jurisdiction and final sign-off authority.

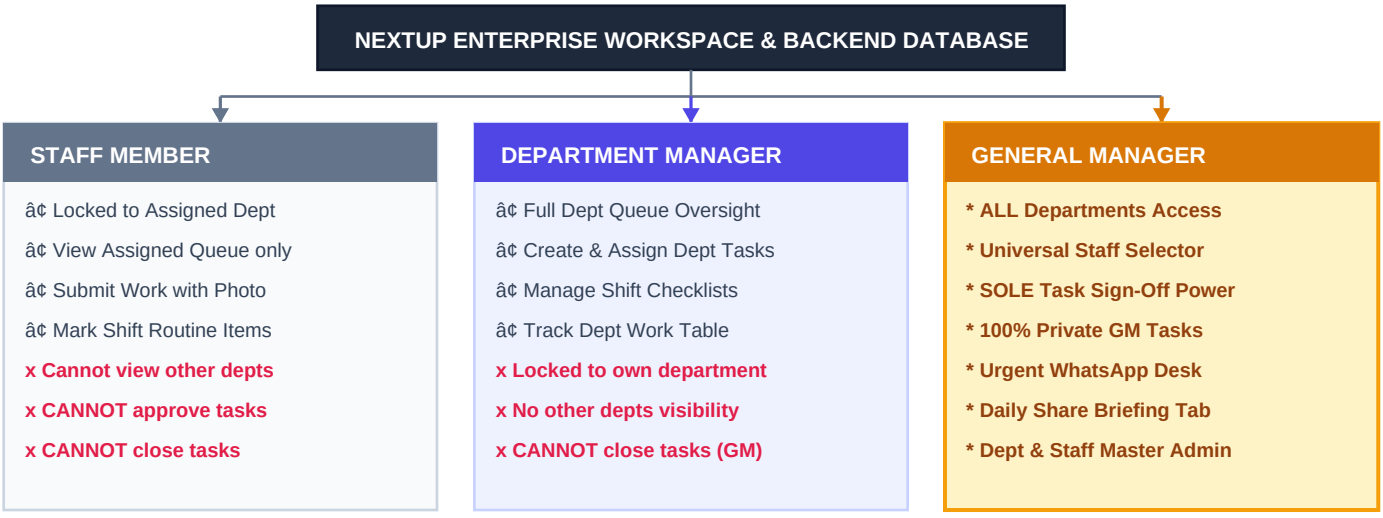
1. PLATFORM SURFACE COMPARISON: MOBILE APP VS WEB PORTAL

Operational Feature	Web Portal (Desktop / Browser)	Mobile Application (Android APK)
Core Operational Purpose	Institutional governance, master data administration, comprehensive work table auditing, and PDF/Excel reporting.	On-the-go shift execution, physical site inspections, photo deliverables via phone camera, and instant WhatsApp dispatch.
Task Workspace Views	4 Powerful Views: Kanban Board (drag-and-drop), Grouped List, Excel-like Table View, and Monthly Calendar View.	2 Touch Views: Modern Touch Cards (default with 16px soft radius) and Mobile Horizontal Table view.
Routine Work Checklists	Desktop shift table (/routine-work) with department tabs, progress meters, completion time logs, and print sheets.	Dedicated Routine Tab in bottom navigation with 1-tap shift checkboxes, progress bar, and instant backend synchronization.
Urgent WhatsApp Dispatch	Direct WhatsApp links embedded on task table rows and urgency lists.	Dedicated WhatsApp Desk Modal with live alert counters, staff groupings, and 1-tap WhatsApp broadcast message composer.
Executive Briefing Share	Copy task link or export custom formatted reports.	Dedicated Share Tab (GM exclusive) to generate pre-formatted executive daily briefings and broadcast to team chats.
Deliverable Proof & Media	File uploader accepting images and files stored on PC / local disk.	Native Camera integration to capture real-time photo proof of work on site, plus phone gallery picker.
Notification Infrastructure	In-app notification bell with unread badges and mark-all-read controls.	Native Mobile Push Notifications with system vibration, sound, lock screen banners, and in-app floating toast alerts.
Staff & Access Management	Full CRUD: Provision new staff, edit credentials, assign roles (STAFF / DEPT_MANAGER), and configure page access.	View institutional directory and filter tasks or checklists by department and individual staff members.
Institutional Reports & Export	Export printable work reports, custom date-range analytics, and CSV / Excel downloads.	Real-time status metric cards (Awaiting GM, To Do, Urgent, In Progress, Done, All Active) with dynamic scoping.
Deployment & Auto-Update	Immediate browser refresh with zero client installation required.	Built-in in-app auto-updater (AppUpdateModal) checking server version on launch with direct APK download link.

2.1 TASK LIFECYCLE & GM SIGN-OFF GOVERNANCE FLOWCHART



2.2 ROLE HIERARCHY & DEPARTMENT ISOLATION ARCHITECTURE



3. GENERAL MANAGER (GM & SUPER ADMIN) EXCLUSIVE AUTHORITIES

GENERAL MANAGER GOVERNANCE NOTICE
To maintain strict organizational integrity, staff members and department managers are locked to their own departments. Only the General Manager and Super Admin possess institutional oversight, universal staff filtering, and exclusive task closeout authority.

1 **Cross-Department Institutional Oversight** GM ONLY
Staff and Department Managers can only view tasks within their designated department. The GM has unrestricted access across all institutional departments (Civil, Mechanical, CSE, Admin, Hostel, Transport, etc.) and can switch views or observe the complete institution in one consolidated stream.

2 **Universal Staff Selector & Individual Work Filtering** GM ONLY
The GM toolbar contains an exclusive Staff Selector button and modal. The GM can filter tasks and routine checklists down to any specific staff member in any department, displaying their exact personal queue and completion performance.

3 **Final Task Review & Official Sign-Off Authority** GM ONLY
When staff complete their deliverables, tasks enter the "Awaiting GM Approval" status (PENDING_GM_APPROVAL). Only the GM has the authority to officially Approve & Close a task, or Request Changes with custom GM feedback instructions.

4 **100% Private "GM Own Tasks" (Executive Workspace)** GM ONLY
The GM has a dedicated "+ GM Own Task" command. These tasks carry an exclusive executive flag (is_gm_own_task: true) and are strictly confidential. They are filtered out completely from all staff and department manager views.

5 **Multi-Department & Cross-Team Task Delegation** GM ONLY
Only the GM can create multi-target tasks assigning a primary deliverable to one department and co-assigning secondary responsibilities to staff members across different institutional wings.

6 **Urgent WhatsApp Dispatch Desk** GM ONLY
The GM mobile header features a live WhatsApp dispatch pill displaying the total count of overdue and urgent deliverables. Tapping it opens a grouped dispatch center to notify staff on WhatsApp with 1 tap.

7 **Executive Daily Briefing Generator (Mobile Share Tab)** GM ONLY
The "Share" tab in the mobile bottom bar is visible strictly to GM and Super Admin users. It automatically compiles daily completion metrics, departmental milestones, and urgent alerts for executive reporting.

8 **Department Provisioning & Manager Designation** GM ONLY
On the Web portal, only the GM and Super Admin can create new institutional departments, modify department codes, assign Department Heads, or archive inactive departments.

9 **Staff Account Administration & Page Security Matrix** GM ONLY
The GM has exclusive control over the Staff Directory: creating login accounts, changing user roles (STAFF vs DEPT_MANAGER), assigning temporary passwords, and managing granular page viewing permissions.

10 **Institutional Analytics Export & SMTP Email Configuration** GM ONLY
Only the GM can configure the institution-wide SMTP email server credentials for task notification emails and export full institutional work report dossiers.

4. ROLE PERMISSION MATRIX: SUMMARY REFERENCE

Platform Capability	Staff Member	Dept Manager	General Manager (GM)
Own Department Tasks	Assigned Only	Full Dept View	Full Institutional View
Other Department Tasks	No Access (Hidden)	No Access (Hidden)	Full Access (All Depts)
Universal Staff Filter	No	Own Dept Staff Only	Yes (All Staff Members)
Task Creation	Self Tasks Only	Own Department	All Depts & Multi-Assign
Final Approval & Closeout	Submit Work Only	Submit Work Only	Exclusive Sign-Off Authority
Private GM Executive Tasks	Hidden	Hidden	Yes (+ GM Own Tasks)
WhatsApp Urgent Dispatch Desk	Receive Only	Receive Only	Send Direct Broadcasts
Daily Executive Share Tab	Hidden	Hidden	Dedicated Mobile Tab
Routine Shift Checklists	Complete Assigned	Manage Dept Shifts	Universal Oversight & Create
Staff & Dept Administration	No Access	No Access	Full Master Management

DOCUMENT GOVERNANCE & AUTHENTICATION

This architecture and authority reference guide reflects the operational state of NextUp v2.3.0. Revisions to role hierarchies or departmental scopes require executive sign-off from the Office of the General Manager.